Assignment Question Paper

Degree	B. Tech.	**Branch**	CSD	**Assignment No.**	II	**Academic Year**	2026-27
Course Code	23CS601			**Max. Marks**	30	**Semester**	V
Course Name	Computer Networks						

Answer any three of the following questions
Each question carries equal Marks

	SET – 1(Roll number 22NU1A4249)			
Q. No.	Questions (1 through 3)	Marks	Learning Outcome (s)	DoK
1 a)	What is the need of flow control? Explain the common approaches for flow control in data link layer?	5M	23CS601.2	L1 – L3
b)	Illustrate framing methods in data link layer with examples	5M	23CS601.2	L1 – L3
2 a) b)	What is the purpose of CSMA with collision detection Write about Standard , Fast and Gigabit Ethernets	5+5M	23CS601.2	L1 – L3
3 a)	Find the minimum Hamming distance for the following cases: i) Correction of 2 errors and ii) Detection of 3 errors or correction of 2 errors.	5M	23CS601.2	L1 – L3
b)	A data word 1011001 is transmitted using even parity. Find the parity bit that should be added to the data. At the receiver, the received data is 1010001. Determine whether a single-bit error has occurred. If yes, identify the error bit.	5M	23CS601.2	L1 – L3
	SET – 2			
Q. No.	Questions (4 through 6)	Marks	Learning Outcome (s)	DoK
4 a) b)	Illustrate the flow control mechanism in data link layer using Sliding Window protocol What is the purpose of CSMA with collision prevention? Explain it	5+5M	23CS601.2	L1 – L3
5 a) b)	Explain Hamming Code Algorithm with Example Assume the user sending the bit sequence **10110011 10101011 01011010 11010101**.Apply the checksum algorithm to check the integrity of the data at sender and receiver side.	5+5M	23CS601.2	L1 – L3
6	Discuss in detail the working of Token Bus and Token Ring. Compare and contrast Token Ring and Ethernet	5+5M	23CS601.2	L1 – L3
	SET – 3			
Q. No.	Questions (7 through 9)	Marks	Learning Outcome (s)	DoK
7 a)	How Hamming distance is related to error detection and error correction. Illustrate with suitable examples	5M	23CS601.2	L1 – L3

7 b)	A bit stream **10011101** is transmitted using the standard CRC method. The generator polynomial is **x^3+1**. Show the actual bit string transmitted. Suppose the third bit from the left is inverted during transmission. Show that this error is detected at the receiver end.	5M	23CS601.2	L1 – L3
8 a)	Explain how Slotted ALOHA solves the problem of channel allocation	5M	23CS601.2	L1 – L3
b)	Describe the stop-and-wait protocol with neat sketch	5M	23CS601.2	L1 – L3
9 a)	What is the significance of data link layer? Explain the design issues of the data link layer	5M	23CS601.2	L1 – L3
b)	Discuss in brief the MAC frame structure for IEEE 802.3	5M	23CS601.2	L1 – L3